The Product

Positioning, a phased roadmap, and the honest boundaries
for AGENTSD / `agentsdaemon.com`

Synthesized from three reviews and prior analysis — Document 2 of 4

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Premise. Two of the three reviews are, at heart, product documents, and they agree on the shape: rebrand to a Unix-daemon identity, lead with an enterprise compliance wedge, and grow into a clearinghouse for machine-to-machine trust. The rigor review is a science document, but it hands the product two gifts it does not know it needs: a defensible security story (channel-level control, not taint hand-waving) and a sequencing discipline (secure the work before pricing it). This document fuses them into one roadmap, and — crucially — marks the claims you may make from each phase, because the fastest way to lose an enterprise deal is to overclaim and get caught.

1 Positioning and the naming question

Verdict. **On the rebrand.** Adopt AGENTSD as the daemon, `agentsctl` as the operator CLI/TUI, `agentsdaemon.com` as the federation. The Unix-service framing (*systemd* manages services, *httpd* serves web, *agentsd* manages the swarm) is genuinely good positioning: it says "background infrastructure," not "another agent framework," which is the correct category to own. Keep "Harbor" as the *unit* noun (a harbor is one principal's daemon-scoped domain) and "Port Daddy" nowhere in the product — it does not survive contact with an enterprise procurement officer. The manuscript can keep whatever internal names it likes; the product needs the clinical one.

The one-line pitch, corrected. The product reviews offer "the Delaware Chancery Court and the Visa network for the machine economy" and "the UCC / POSIX standard for AI labor." These are good *aspirational* lines for the clearinghouse endgame, but they are Phase-5 claims worn on a Phase-1 body. The rigor review supplies the honest Phase-1 line, which is stronger because it is deliverable: closest to **Lloyd's for autonomous work** — standard work orders, declared risks, evidence, and (eventually) underwriters — built on a local **control plane and flight recorder** for AI agents. Lead with the flight recorder; grow into Lloyd's; mention Visa only once you clear funds.

2 Go-to-market: the three stages, re-sequenced and re-claimed

The product reviews' three-stage moat (Enterprise Compliance → B2B Airlock → Harbor Economy) is the right arc. The correction is what you are *allowed to say* at each stage.

Stage	What it is	Claims you may / may not make
1. The Wedge *Compliance & liability containment*	Sell AGENTSD as an API firewall and flight recorder: cryptographic audit logs, hard budget firewalls (SIGKILL on breach), OS-level effect confinement, and a *work receipt* for every task saying what was verified vs. merely observed. Buyer: the enterprise terrified its agents will leak keys or drop a database.	**May:** "complete, tamper-evident audit log of every mediated effect"; "hard budget caps enforced at the SQL engine"; "no ambient network/credential access outside brokered channels." **May not:** "cannot exfiltrate" (only "cannot exfiltrate through mediated channels; here is the enumerated channel set"). Ship the assurance-level label (below) with every deployment.
2. The Airlock *Silicon-enforced NDAs / clean room*	The Derek/Erin product: run Erin's encrypted agent against Derek's encrypted data in an attested confidential VM, dual-key release, two declassification gates, priced leakage budget. Platform fee for facilitating the cryptographic DMZ.	**May:** "every release is explicit, gated, and metered against a declared leakage budget"; "neither party's key is released except to the attested measurement." **May not:** "mathematically cannot phone home" (replace with the budget claim). The honest version is a *better* sale to a CISO who has seen a side-channel talk.
3. The Clearinghouse *Settlement, admission, arbitration*	Charge settlement fees on float plans (you provide oracle-bound closure), harbor-admission fees (Sybil deterrence), and arbitration fees for disputes. The 2-of-3 multisig escrow keeps you non-custodial.	**May:** "non-custodial 2-of-3 settlement; the platform never takes custody on the happy path"; "bonded, rated arbitration." **May not:** "trustless" (you are a cryptographic *judge*, which requires a trusted adjudication root) or any yield/return language until counsel clears MSB/regulatory exposure.

On neo-feudalism. The product reviews' "knights wearing colors, lords posting bonds, invitation-only as a feature" is vivid and correct as an *account of why high-trust networks are exclusionary* (the bonded-sponsor pattern). Use it in a blog post and a keynote. Do *not* put it in the security documentation, where "exclusionary by design" reads as "we could not solve open admission" — which the rigor review shows is a real, priced trade-off (sponsor concentration caps, sponsorship auctions, automatic probation expansion), not a virtue to celebrate.

3 The assurance-level label: your most valuable product artifact

The rigor review's five levels are not academic hedging — they are the single feature that lets you sell honestly and defensibly. Ship this label on every deployment and every marketing claim:

1. **Observed** — effects may occur outside the daemon; evidence is best-effort. (Hook-based integrations start here.)
2. **Coordinated** — well-behaved clients use claims, roles, commitments.
3. **Brokered** — credentials and selected effects pass through daemon-owned brokers.
4. **Confined** — the process has no ambient route around fs/network/tool/credential brokers.

5. **Attested** — a remote party can verify the confined runtime and policy measurement.

The product reviews' entire pitch presumes level 4–5 ("mathematically cannot"), but their default integration path — IDE extension intercepting MCP calls, LLM API proxy, PATH shims — is level 2–3 (a coordinator, bypassable by a determined same-user process). *Do not claim level 4 while shipping level 2.* The hypervisor work (Phase 2) is what earns levels 4–5, and only "for the enumerated channel set" after a red team has hunted the negative space (raw sockets, DNS, `/proc`, inherited descriptors, git helpers, package managers, child processes, clipboard, provider callbacks).

4 The phased roadmap

Sequenced so each phase ships a claim the previous phase earned. The unifying discipline (from the rigor review): *secure the work, then preserve and share it, only then price it.*

Phase 0 — Repair the spine (weeks, not months)

Before any new machinery: land the eight consensus corrections from Document 1 (SDT direction, exact-key exclusion + fencing epoch, identity theorem, float-plan buckets, Anchor edge check, WAL durability types, schema migration ledger, and the "cryptography $\neq$ reality" scoping). These are cheap and they are the difference between a system a security engineer trusts and one they do not. Ship the assurance-level label and a per-component threat-model table.

Phase 1 — The one-agent receipt (the wedge MVP)

The smallest product that proves the thesis, at n=1:

- the work-unit object and lifecycle; a typed *acceptance policy* and completion *receipt*;
- an artifact/evidence graph answering why / where / why-not for every claim;
- structured live-state projection (obligations, grants, pending effects, claim verdicts);
- a portable task capsule with *same-provider* restart under a minute (event-sourced message array + Git SHA + prefix-cache replay — the product reviews' mechanism, correctly scoped);
- an operation journal giving tool idempotency (at-most-once external effects);
- atomic mutual exclusion on the enforced conflict predicate; oracle-bound closure that rejects free-text "all green, trust me."

Ships the claim: "a successor can safely continue after termination, and a human can audit why *done* was accepted." That is the compliance wedge. **Distribution:** pre-compiled binaries (`npm/pip/brew`) and the IDE extension that boots the daemon silently and intercepts MCP calls — the product reviews are right that vibe-coders will not install a Rust toolchain. Label this integration **Coordinated/Brokered (level 2–3)** honestly.

Phase 2 — Roles, typed messaging, confined effects (earn level 4)

- durable *role* mailboxes addressed by role/topic (`role:test-owner`, not `agent-492`) so the mailbox survives resurrection; typed performatives (INFORM, REQUEST, PROPOSE, COMMIT, ACCEPT, REFUSE, CHALLENGE, WARN, EVIDENCE, RETRACT) with explicit deontic semantics — INFORM creates a claim not a truth, a message never carries tool authority because its text asks for an effect;
- sole-responsibility *role leases* with fencing epochs (calendar avatar, test writer, roadmap owner) — one accepted epoch per exclusive role, verification kept separate from ownership (a test-writer role must not be the sole approver of its own tests);

- the effect hypervisor: microVM/VM isolation, seccomp/Landlock/gVisor confinement, cgroup quotas, a network-policy proxy, credential broker (short-lived scoped secrets), tool broker (calendar/GitHub/cloud/payment), transactional outbox;
- the SLM sidecar as an *untrusted projection service* — lower privilege than the daemon, no effect authority, agent text as tainted data, provenance required per generated claim, mandatory-obligation IDs computed *outside* the model. Name it "evidence compactor," not "memory."

Ships the claim (red-team gated): "Confined (level 4) for the enumerated channel set." This is where the psychosis-detection loop lives too — but framed correctly (below).

Phase 3 — The review fleet and prospective assurance (the demand generator)

Build the GitHub review fleet and Lookout/Unspider on the same claim/effect graph. The rigor review's key insight: do not have five agents read the same diff and vote (correlated failure). Use *deliberately different information* — a property-author who writes tests from the spec *before* seeing the patch, a blind adversarial tester, a security/effect reviewer, a merge-risk reviewer, a protected-CI adjudicator emitting signed receipts. Adversarial test-writing is scored by *mutation kill rate* on a held-out mutation set. This provides adversarial *demand* for evidence and generates the incident corpus that CDM skill-extraction needs. **Ships:** "assurance as a purchasable line item" — $(1-d)^k$ at a stated price — which is the clearinghouse's core value made concrete two phases early.

Phase 4 — Provider-portable resurrection (the flagship demo)

Publish the neutral task-capsule schema; write provider adapters that reconstruct context but never pretend to restore hidden inference state. Demonstrate the money shot: Claude credits exhausted mid-task, a different provider resumes, no external effect duplicated. **Names its own attack:** engine substitution (build reputation on a strong model, spend it on a weak one) — defend with engine attestation in every witnessed outcome, reputation indexed by (principal, engine). **Ships:** the "big resume button," the single feature that makes the continuity story legible to a buyer in five seconds. Say *task continuity with actor replacement*, never "perfect neural continuity."

Phase 5 — The hosted project institution (Alice, Bob, Carl)

Start with Alice and Bob under one hosted authority for the non-monotone operations (exclusive roles, revocation, finite budgets, unique settlement); keep personal state in personal harbors. Add the coordination graph (typed modalities: observed/believed/desired/intended/committed/required/perm typed-conflict detection (deterministic resource checker | SMT for invariants | temporal-network for schedules | policy block for security | steward decision for UX), the durable agentic estate (actor build / instance / memory / relationship / authority, separated), and product-journey traceability. Add Carl only after two-person semantics are stable (the rigor review is right that n=3 adds coalitions, which is why the charter votes on charter, never on truth or another's namespace). **This is the remote clearinghouse's real substrate**, and it is more valuable than the shared repo because it explains why the repo exists.

Phase 6 — The Sealed Harbor (the airlock, hardened)

Begin with a no-network microVM whose rich result only Derek receives; add dual attested key release, measured policy, whole-worker taint, fixed output channels, signed receipts, and adversarial exfiltration red-teaming (canary bits through network, logs, errors, filenames, length, timing, CPU load, receipt status). Confidential-GPU attestation and special MPC/FHE kernels for narrow fixed computations come later. **Ships the corrected airlock claim:** bounded, declared, gated release — not magical secrecy.

Phase 7 — Economy and underwriting (price it, finally)

Only now, with real receipts and real failure rates, does the market price them. Bind bonds, provider collateral, insurer coverage, and dispute procedures to principal-owned work orders and evidence. Model correlated failure by provider, model family, toolchain, and shared dependency (the rigor review's repeated warning: do not sum risks as if model/provider/dependency failures were independent). Include buyer misconduct and evidence poisoning in the threat model. The 2-of-3 multisig escrow (product reviews' best financial idea) is the settlement rail; the underwriting layer prices the residual after prevent/contain/checkpoint/observe/adjudicate.

5 The feature adjudications: keep, fix, or cut

The product reviews propose a dozen concrete features. Verdicts, with the fix where the intent is good but the mechanism is wrong.

Feature	Verdict	Reasoning / required fix
2-of-3 multisig escrow	**Keep**	The best financial idea in either review. Non-custodial on the happy path (Alice+Bob sign), arbiter's key only for disputes. Solves the MSB-honeypot problem the manuscript's "recipient-whitelisted custody ledger" created. Ship it; get counsel on the arbiter role.
Event-sourced rollback / "swarm time travel"	**Keep, rescope**	The mechanism (truncate the message array to a timestamp, `git reset`, replay under prefix cache) is the right capsule implementation. Drop "perfect neural continuity"; it is same-provider behavioral continuity, cross-provider succession.
Asylum protocol (psychosis detection)	**Keep, rename**	Detecting "5,000 tokens, zero AST progress" via a causal-density ratio is a real and useful fault class. `SIGSTOP` → excise loop → inject corrective prompt → `SIGCONT` is sound. Rename "Vibe Time" to a formal term (the rigor review's *Causal Work Time* plus a separate *hazard* clock; four clocks, not one) and keep "the Asylum" as internal slang.
2-of-3 escrow arbiter as "cryptographic judge"	**Keep, don't call it trustless**	Correct design; honest framing. You are a trusted adjudication root with cryptographic settlement, not a trustless protocol.
Darwinian prompts (skill Elo)	**Keep, harden**	Awarding skill Elo on settled success is good, but naive Elo is gameable and off-policy. Score on *witnessed downstream outcome*, warm-start new skill versions via provenance-weighted inheritance with a no-mint conservation bound (Document 3, A6), and hold out evaluation tasks to blunt Goodhart.

Feature	Verdict	Reasoning / required fix
Selective checkpointing (FULL sync for money)	**Keep**	Exactly right: `NORMAL` for ephemeral claims, `wal_checkpoint(FULL)` before returning on money-bearing writes. This *is* the per-write durability type from Document 1, Change 7.
SQL-level overdraft (`UPDATE...WHERE bal>=b RETURNING`)	**Keep**	Correct fix for the async-tick race; enforce exclusion in the engine, not the app layer. Generalize to the conflict predicate for claims.
Differential fuzzing for runtime parity	**Keep, relabel**	1M random ops against two bindings with `hash(dump(db))` equality is good *conformance evidence*; it is not a proof of I11 (the rigor review is right). Label it conformance testing in CI, not a theorem, and include the deployed binary.
Dynamic taint analysis for the airlock	**Demote to a layer**	Useful as the *contain* rung, but token-level taint through an LLM is not soundly definable and "cannot phone home" is an overclaim. The real boundary is the declassification gate with an information-theoretic budget (Document 1, Adjudication 3).
"JSON ledger IS the neural state"	**Cut the claim**	Keep the ledger; delete the identity claim. Two hidden states can share a transcript (Document 1, Adjudication 2).
Inception Canaries (inject real vuln, score operator)	**Fix or cut**	Intent good (fight abdication), mechanism unacceptable: deceiving your operator and writing a live vulnerability that can escape the harness. Replace with *consented, labelled* probes from a blinded holdout of already-fixed defects replayed in a sandbox (Document 1, Adjudication 5).
Inbox-injection via "top-level system mutation"	**Fix**	Delivering operator interrupts by writing to the model's system channel so it treats them as "un-challengeable ground truth" is precisely the prompt-injection attack surface the rigor review warns about. Deliver interrupts as *typed, provenance-labelled* control messages the runtime renders as data, with an out-of-band operator override that is higher-priority than agent work and coupled to effect revocation — not as forged ground truth.
Uncapped slashing for Sybil deterrence	**Keep, with principal binding**	Removing the `min(B_dep, B_T)` cap so a $1,000 stake is fully slashed for sabotaging a $100 task does raise attack cost — but only bites when bonds bind to a *principal* who cannot cheaply re-enter. Pair uncapped slashing with principal-aggregated exposure and onboarding cost, or a Sybil just mints a fresh staker.

6 What the product is, in one paragraph

AGENTSD is the control plane and flight recorder for AI agents: a local single-writer daemon that mediates every consequential effect, records an auditable evidence-bearing receipt for every unit of work, confines agents to brokered channels, and can resume a task across a provider change without duplicating an external effect. It sells first as compliance and liability containment (the wedge), then as a mutually-confidential clean room for parties who cannot share data or models (the airlock), then as a non-custodial clearinghouse that settles and underwrites machine-to-machine work (the economy). Every claim it makes is labelled with an assurance level it can defend, and every release it permits is explicit and bounded. That is a product a CISO can buy and a regulator can tolerate — which is worth more than a product that promises magic and gets breached.